Arista Access Node HEB-ACC-25 Standard Operating Procedure (SOP) for BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding

Node: HEB-ACC-25 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5999

Description

This SOP outlines the procedures to investigate, diagnose, and resolve Routing Protocol Flap failures on the Arista Access Node HEB-ACC-25 at the Hebbal Edge site. The failure domain is characterized by BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This scenario typically occurs due to misconfigurations, hardware issues, or software bugs in the routing protocol stacks. The Arista EOS (Extensible Operating System) manages the routing protocols on this node, and the CLI is used to configure and troubleshoot these protocols. The underlying mechanics of this failure domain involve the interaction between the routing protocols and the underlying hardware, including the switching and forwarding engines. In this scenario, the BGP route churn may be caused by an incorrect BGP neighbor configuration or an issue with the BGP MIB (Management Information Base). OSPF adjacency loss may be due to an incorrect OSPF configuration, an issue with the OSPF MIB, or a hardware problem. ISIS LSP flooding can be caused by a misconfigured ISIS LSP (Label Switched Path) or an issue with the ISIS MIB. The failure of these protocols can lead to network instability and packet loss, impacting the overall network availability and performance.

Pre-requisite Checks

- Step: Check the power and network connectivity of the node, Description: Verify that the node has power and is connected to the network

- Step: Verify the redundant link status, Description: Check if the redundant link is up and functioning correctly

- Step: Check the BGP MIB, Description: Verify the BGP MIB configuration and ensure it is correct

- Step: Check the OSPF MIB, Description: Verify the OSPF MIB configuration and ensure it is correct

- Step: Check the ISIS MIB, Description: Verify the ISIS MIB configuration and ensure it is correct

Potential Root Causes

Cause	Probability
BGP misconfiguration	High
OSPF misconfiguration	Medium
ISIS LSP misconfiguration	Medium
Hardware issue	Low
Software bug	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays BGP summary information, including the number of peers, the number of routes, and the number of prefixes. It can help identify BGP-related issues.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays OSPF neighbor information, including the neighbor's IP address, the neighbor's state, and the neighbor's interface. It can help identify OSPF-related issues.

Command: show isis adjacency, Expected Output: ISIS adjacency information, Description: This command displays ISIS adjacency information, including the neighbor's IP address, the neighbor's state, and the neighbor's interface. It can help identify ISIS-related issues.
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Resolution Steps

Step 1: Step: Correct the BGP misconfiguration, Command: config t router bgp 64512 neighbor 10.10.10.1 remote-as 64512, Description: Correct the BGP misconfiguration by updating the BGP neighbor configuration.

Step 2: Step: Correct the OSPF misconfiguration, Command: config t router ospf 1 network 10.10.10.0/24 area 0, Description: Correct the OSPF misconfiguration by updating the OSPF network configuration.

Step 3: Step: Correct the ISIS LSP misconfiguration, Command: config t router isis level-1-2 10.10.10.0/24, Description: Correct the ISIS LSP misconfiguration by updating the ISIS LSP configuration.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP summary information, Description: Verify that the BGP summary information is correct after correcting the BGP misconfiguration.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: Verify that the OSPF neighbor information is correct after correcting the OSPF misconfiguration.

Command: show isis adjacency, Expected Output: ISIS adjacency information, Description: Verify that the ISIS adjacency information is correct after correcting the ISIS LSP misconfiguration.
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Roll-back Procedures

- Step: Restore the previous BGP configuration, Command: config t router bgp 64512 show running-config | include router bgp 64512, Description: Roll back to the previous BGP configuration if the fix fails.

- Step: Restore the previous OSPF configuration, Command: config t router ospf 1 show running-config | include router ospf 1, Description: Roll back to the previous OSPF configuration if the fix fails.

- Step: Restore the previous ISIS LSP configuration, Command: config t router isis show running-config | include router isis, Description: Roll back to the previous ISIS LSP configuration if the fix fails.